

Lec 1

Knowledge and Understanding:

- 1.1 Social harms enabled by computing
- 1.2 Professional codes of ethics
- 1.3 Professional responsibilities
- 1.4 Equity-Diversity-Inclusion in computing
- 1.5 Privacy and data protection
- 1.6 Ethical challenges of contemporary computing
- 1.7 Responsible research and innovation

Intellectual Skills:

- 2.1 Understanding the nature of ethics in computing

Professional Skills:

- 3.1 Knowledge, understanding and application of ethical codes in computing

What is **Professional Ethics in Computing**?

1. **Computing:** Hardware, Network, Algorithm, Media, Data, Communication
2. **Ethics:**
 - (1) Human beings need to socialize and exchange products and information; ethics provides social convention to facilitate this.
 - (2) Determine what is right and wrong.
3. **Professional:**
 - (1) Competence in skilled work.
 - (2) Certification
 - (3) Agency: make work decisions that may require ethical judgement.

Ethical problems in computing often arise because technical decisions have social consequences.

1. **Computing systems can enable significant social harms.**
2. **Professionals have agency and therefore moral responsibility.**
3. **Ethical issues arise when technical decisions affect people.**
4. **Efficiency and fairness are often in tension.**
5. **Ethics provides frameworks to reason about difficult choices.**

Lec 2

Definitions: Ethics, Morality and law

1. **Ethics:** Set of principles of how each member of a group ought to behave, shared by the group.
2. **Morality:** One's individual belief about how one ought to behave. This typically implies an expectation of others.
3. **Law:** Social rules of right and wrong, enforced by the state (i.e. police and law courts).

Difference between “is”(是) and “ought”(应该) :

- | | |
|-------------------------------------|--------|
| (1) Mr. Smith is paying his tax. | 他在缴纳税 |
| (2) Mr. Smith ought to pay his tax. | 他应该缴纳税 |

Relations: Ethics, Morality and Law

These three ideas overlap but are not necessarily the same. 三者可重叠但不等同

Some interesting ideas about Ethics:

1. Ethical decisions may differ between different groups (Relativism)

- (1) Ethics depends on the group which is being considered

For this module: listen carefully, respect other viewpoints, do not personalize discussions.

2. Context

- (1) Ethical decisions often depend on context.

Lessons:

- (1) Sometimes an ethical decision cannot be made until more information is gathered.
- (2) Sometimes a poor ethical decision is made on misinformation.
- (3) Sometimes an ethical decision needs to be changed in the light of new information.

3. Cost is important

- (1) The consequence or “cost” of ethical decisions is important.

Lessons:

- (1) Actions have different unethical “degrees” (not just true/false).
- (2) Should we (can we) “cost” outcomes quantitatively?

Ethical decisions often depend on context and the potential consequences of actions.

4. Egoism versus altruism

Two distinct philosophical viewpoints:

(1) Egoism:

- (1) Everyone is motivated to do only what is good for themselves and individuals
- (2) Hence, only ethical theory is "Do what is best for me"
- (3) Clearly, this is problematic: Selfish and uncaring society

(2) Altruism:

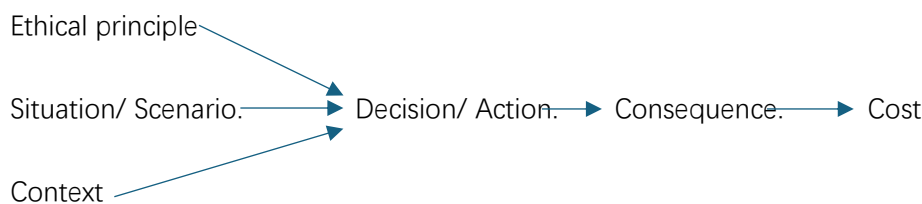
- (1) Ethical principle: Do what is good for other people
- (2) Altruism may not be realistic!
- (3) At its extreme, this is not good because people may not care for themselves

Modern approach blends egoism and altruism:

- (1) Following social rules (altruism) is good for individual (egoism)
- (2) Caring for self (egoism) is a requirement before helping others (altruism)

5. Framework for Ethical Decision-making

To help us discuss ethical decision-making we can therefore provide the following general framework:



Theories of Ethics:

1. Cultural relativism

- (1) Ethical principle: An action is judged good or bad based only on the standards adopted by one's society
- (2) This follows from observation that ethics is different between groups
- (3) No valid rational criterion for determining the right thing to do exists
- (4) Different cultures cannot be judged against each other
- (5) Positives: Promotes tolerance
- (6) Negatives: Difficult to defend

Pluralism:

- (1) Cultural relativism is problematic because it is not objective: we cannot apply the same ethical standards to people from different cultures.

- (2) We need a universal ethics code for the computing profession

However, two attributes of culture appear to be universal amongst all humans: language and morality, there is a "natural morality".

- (3) However, for us on this module, the case for cultural relativism is weak. It does not provide clear guidance in deciding what is right when interacting with people of other cultures.

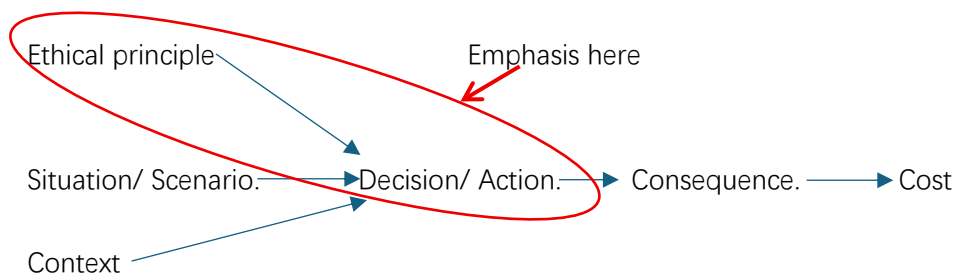
2. Categorical imperative

- (1) **Ethical principle: An action is moral if it can be stated as a universal rule without contradiction**
- (2) This derives ethics directly from reason. The process is called the “categorical imperative”.
- (3) A flaw with the categorical imperative is that it forms universal laws that must be logically true.
- (4) No room for exceptions.

From a deontological perspective, the action is morally wrong because it violates a universal duty, regardless of its consequences. (考试标准句)

3. Deontological Ethics

- (1) Categorical imperative is a type of Deontological Ethics
- (2) Concerned with what people do, not with consequences of actions
- (3) Do the right thing because it is your duty

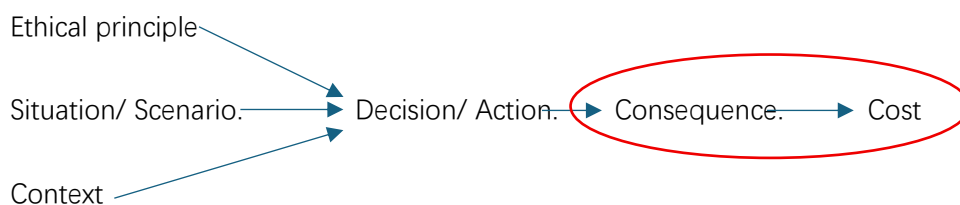


Good intentions:

- (1) Deontological Ethics focusses on **the intention of an action**
- (2) Principle of “good intention”
- (3) But if an ethical decision is badly informed then it may lead to bad consequences

4. Utilitarianism

- (1) Hedonism is the principle that actions that make me happy are the best. However, this is egoistic.
- (2) **Utilitarianism is the altruistic development of hedonism**
The right decision is the one that causes the most happiness, amongst all people affected by the decision.
- (3) Utilitarianism is a reaction to Deontological Ethics focusing on the consequence.
- (4) Hence, it is a form of Consequentialism: focus on outcome, not intention.



Maximize happiness (consequence)

What is happiness?

(1) Physical, emotional, mental:

- (1) Satisfaction of accomplishment, appreciation of interesting idea, aesthetic pleasure of art, pleasure of helping others, ...
- (2) Absence from pain and suffering.
- (3) Sometimes "happiness" is called "utility" (usefulness).

However, utilitarianism requires a way to quantify or "cost" the consequence.

This is not very easy and may lead to crude oversimplification. It may not even be possible.

Utilitarianism has a worse problem: **tyranny of the majority (多数暴政)**

Because utilitarianism seeks to maximize total happiness amongst all affected people, then improving happiness at the expense of a minority is ethically acceptable, according to utilitarianism. (为了追求最大化幸福, 牺牲少部分人的幸福是可以的)

While utilitarianism may justify this decision due to overall benefits, it can be criticized for disproportionately harming a minority group. (万能考试句)

5. Contractarianism

- (1) Both a political and a moral theory
- (2) Can be merged with consequentialist, or deontological theories.
- (3) Premise:
 - (1) People are rational and wish to promote self interests (egoism).
 - (2) The best way of doing that is to promote the common interests of society.
- (4) Ethical decisions are those that are in accordance with common interests, the social contact.

From a contractarian perspective, ethical actions are those that uphold the implicit social contract between professionals, users, and society. (考试万能句)

6. Ethics of Care

- (1) A single moral ideal virtue: **caring for other individuals.**
- (2) Goodness is whether or not we take care of people around us.
- (3) One-caring: someone who fulfils the role of a care giver.
- (4) To act morally, we need to make ourselves one caring for those with whom we have relations.

Components of Caring:

- (1) Caring is characterized by feeling of engrossment in the needs of another person and by acting to care for that person.
- (2) For caring to happen, it is necessary for the cared for person to be receptive.
- (3) We can "care about" everyone, but we cannot "care for" everyone.
Hence, it is morally acceptable (and necessary) to only care for a small number of

individuals

(4) Since caring involves the specific situation of the specific individual, it does not lead to universal moral rules:

(1) Focus on specific cases and analogy, not reasoning.

Caring by Computer Professionals:

(1) It's not possible to "care for" all (hundreds or millions of) users of a software application; but software can help people.

The ethics of care criticizes this action for failing to attend to the needs of those most vulnerable. (考试万能句)

1. Ethical decisions cannot be reduced to personal opinion.
2. Different ethical theories prioritize different moral values.
3. Utilitarianism focuses on consequences, while deontology focuses on duties.
4. Ethical decisions often involve trade-offs between competing values.
5. Treating everyone the same does not necessarily result in fairness.
6. Caring ethics emphasizes vulnerability and relationships rather than rules.
7. While a utilitarian approach may justify the system due to overall benefits, a deontological perspective would criticize it for violating individual rights, regardless of outcomes.

简答题的回答流程:

Scenario → Action → Consequence → Cost → Principle → Context

Lec 3

Impact of Computing:

- (1) Ethics is needed to ensure that we deploy computer systems that are a positive experience for people and **not harmful**.
- (2) Software developers and computer engineers have a **responsibility** to ensure ethics in computing.

Even systems that are not safety-critical can have serious negative impacts on people.
(考试万能句)

Some common ethical principles in Computing:

- (1) Contribute positively to society and human well-being.
- (2) Be honest, transparent and trustworthy.
- (3) Maintain professional competence for yourself and other coworkers.
- (4) Respect data privacy and intellectual property rights.
- (5) Be fair, do not discriminate, and take action against discrimination.

易考点:

公平 (fairness) ≠ 一视同仁

透明 (transparency) ≠ 解释一切

Ethical dilemmas and critical reasoning

- (1) **Use your judgement** and reinforce with theories of ethics.
- (2) **Use professional codes of conduct**, as source of guidance:
 1. ACM/IEEE Software Engineering Code of Ethics and Practice
 2. British Computer Society (BCS) Code of Conduct

总结为考试万能句:

Use your judgement, reinforced by ethical theories and professional codes.

Key principles in the BCS Code of Conduct:

- (1) Public interest
- (2) Professional competence and integrity
- (3) Duty to relevant authority
- (4) Duty to the profession

Why don't we always do the "right" thing?

- (1) We may do the wrong thing by **mistake or ignorance** (e.g. introduce a bug.)
(通常 → forgivable error)

- (2) **Negligence(疏忽)** **negligence ≠ accident**
- (3) We may do the wrong thing because we **are intentionally bad**.
 This constitutes a serious ethical breach.
- (4) **Relativism.** We may not agree on what is “right”.
 不构成正当理由
- (5) **Coercion.** People may be deliberately forced to do something unethical, (but in modern world it will be more subtle: peer pressure, or managerial pressure, withholding of benefits, blackmail and so on.)
- (6) People are more **complex** than simply “good” and “evil”!

- 1. **Computing systems can cause significant social harm.**
- 2. **Software developers have professional responsibilities beyond writing correct code.**
- 3. **Ignorance may explain an error but does not always excuse it.**
- 4. **Ethical responsibility is often shared across organizations and individuals.**
- 5. **Ethical judgement should be supported by theories and professional codes.**

Lec 4

What is a profession?

- (1) medical doctors,
- (2) lawyers,
- (3) engineers,
- (4) teachers,
- (5) nurses,
- (6) accountants,
- (7) architects,
- (8) and many others.

Characteristics of professions(Bayle):

- (1) Specialized Knowledge
- (2) Formal Education and Training
- (3) Ethical Standards
- (4) Certification or Licensing
- (5) Professional Associations
- (6) Accountability
- (7) Continuing Education

六个核心特征(写 5-6 个即可):

- (1) Provides an important service to society
- (2) Requires extensive training
- (3) Involves significant intellectual effort
- (4) Organisation of members
- (5) Individual autonomy
- (6) Certification or licensing

考试万能句:

According to Bayles, a profession is characterised by specialised knowledge, extensive training, autonomy, and a commitment to serving society.

What are professional ethics?

A set of morally permissible standards of **a group that each member of the group** (at his/her rational best) wants every other member to **follow** even if their doing so would mean that he/she must do the same.

总结为:

A set of morally permissible standards of a group that each member wants every other member to follow, even if it means they must do the same.

但你要抓住三点:

- (1) group-based (群体标准)
- (2) shared expectations (共同期望)
- (3) reciprocal obligation (互相约束)

Professional ethics - A subset of moral concerns

- (1) Morality is much broader in scope than professional ethics

- (2) Morality encompasses societal reasoning and norms of conduct as to what constitutes right and wrong
- (3) Professional ethics govern professional practice with respect to particular moral issues or challenges, e.g., algorithmic decisions
- (4) As the broader social-moral order evolves so do professional ethics (ethics a matter of life-long learning)

General ethics vs. professional ethics

- (1) It is against ethical standards in the USA for doctors to advertise prices for their services, but there is nothing inherently immoral about advertising prices for services
- (2) This means that an action may be morally permissible but contravene professional professional standards
- (3) Professional ethics requires that one behaves consistently with the standards of the group

对比考点：

- (1) **General ethics:** 适用于所有人；更宽泛；基于个人判断
- (2) **Professional ethics:** 适用于专业群体；更严格；基于群体标准

关键总结:

An action may be morally permissible in general, but unacceptable under professional ethics.

Professional standards consist of:

- (1) Principles
- (2) Rules of conduct
- (3) Embodied in codes of conduct or codes of ethics

What constitutes 'professional activity'?

- (1) Provides an important service to society
- (2) Requires extensive training
- (3) Involves significant intellectual effort
- (4) Organisation of members
- (5) Individual autonomy
- (6) Certification or licensing

Is computing a profession?

- (1) Lack of accreditation, certification or licensing

Computing ≠ traditional profession

严格意义;

- (2) No single organisation of members for the computing profession

- (3) But:

- (1) The harms enabled by computing mean that people doing computing as a job of work still have important ethical obligations

- (2) "Programmers need ethics when designing the technologies that influence people's lives." ----President of the Association of Computing Machinery

总结为:

Despite this, the harms enabled by computing mean that computing practitioners still have serious ethical obligations.

考试万能句:

While computing may not fully meet the traditional criteria of a profession, those working in computing still bear professional ethical responsibilities due to the significant social harms their work can enable.

Disadvantages of professions:

- (1) Economic Inequality
- (2) Access to Services
- (3) Professional Monopolies
- (4) Bureaucracy and Red Tape
- (5) Resistance to Change
- (6) Stress on Public Resources
- (7) Overemphasis on Credentialing
- (8) Lack of Diversity
- (9) Professional Ego and Entitlement

Three codes of ethics:

- (1) BCS
- (2) ACM
- (3) IEEE

Code of ethics – 6 key areas of professional responsibility

- (1) **General expectations** of professional conduct
 - (1) You are generally expected to uphold the profession
 - (2) Unethical conduct includes:
 - (1) Exaggerating skills and competences
 - (2) Withholding or misrepresenting technical information
 - (3) Conflicts of interest
 - (4) Divulging confidential information
 - (5) Dishonest conduct
- (2) Exercise **due diligence** in professional work
 - (1) It is imperative that you **take reasonable care in conducting your professional work**
 - (2) This means:
 - (1) You are competent to do the work required of you and your team

- (2) Appropriate steps are taken to avoid harm
- (3) Taking reasonable care
- (4) Also means:
 - (1) Systems are robust, secure and respect privacy
 - (2) Special care is taken when modifying or retiring systems or when systems are integrated into societal infrastructure

经典考试结论:

Failure to exercise due diligence constitutes negligence rather than a forgivable error.

(3) **Avoid discrimination** (in the workplace and systems design and use)

- (1) In workplace or design and use of systems
- (2) A specific form of harm:
 - (1) Based on **personal characteristics** (gender, race, age, disability, religion, socioeconomic status, etc.)
 - (2) **Harassment, abuse, oppression**
 - (3) **Examples of discriminatory systems**

考试万能句:

Algorithmic systems can unintentionally reinforce existing forms of discrimination if fairness is not carefully considered.

(4) **Respect intellectual and intangible property**

- (1) **Respect the work of others**
- (2) Two key aspects:
 - (1) **Do no harm**
 - (2) **Do not hack:** Unless public good requires it or you are authorised to do so
 - (3) **Respect intellectual property rights (IPR)**
 - (4) **Limits of protecting your own work** (e.g., Sony rootkit)

(5) Always act in the **public good**

- (1) Common to BCS, IEEE and ACM codes
- (2) Central to professional conduct:

“Computing professionals’ actions change the world. To act responsibly, they should reflect upon the wider impacts of their work, consistently supporting the public good.”

考试结论:

Acting in the public good is central to professional conduct, even when it conflicts with organisational interests. (非常适合 whistleblowing 题目)

(6) **Whistleblowing**

- (1) The final recourse
 - (1) Ethical codes are all about your conduct.
 - (2) Incumbent on you to challenge unethical conduct.
 - (3) Negligence by inaction not ethical.
 - (4) Whistleblowing, legislation and public interest.

明确立场:

- (1) Ethical codes concern your conduct**
- (2) Silence ≠ neutral**
- (3) Negligence by inaction is not ethical**
- (4) Whistleblowing = last resort**

考试必写点:

- (1) public interest
- (2) proportionality
- (3) internal channels first
- (4) legal protection

标准句:

When serious harm to the public is foreseeable, whistleblowing may be ethically justified as a last resort.

考试万能句:

- (1) Professional ethics impose stricter standards than general morality.**
- (2) Computing may not be a traditional profession, but it enables significant social harm.**
- (3) Due diligence is a core professional responsibility.**
- (4) Acting in the public good may conflict with organisational interests.**
- (5) Negligence by inaction is not ethically acceptable.**

Lec 5

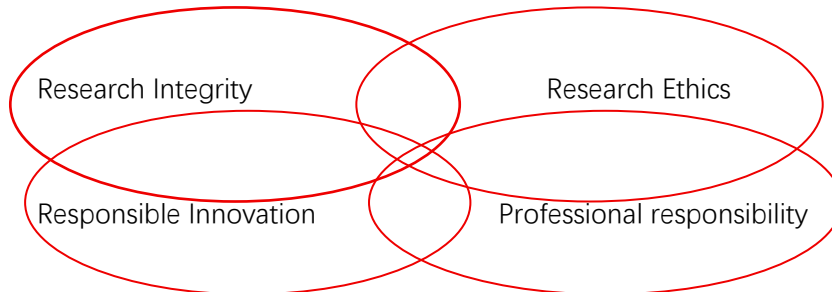
Research in Computer Science

- (1) Research allows us to **understand the world better and develop new technology.**
- (2) The field of Computer Science is **developing very fast** with **many new innovations** in recent years.
- (3) Research conducted in **University**.
- (4) Research also conducted increasingly in **Industry**:
 - (1) Google Deepmind, Microsoft OpenAI, Baidu Research...
- (5) Even in a smaller firm, a computer programmer may need to do some research:**
 - (1) e.g. Optimal scheduling system for van drivers in a medium-size business.

总结为:

Research ethics ≠ 只属于“科学家”. 而是 所有 computing professionals 都可能涉及
Research ethics is not just for "scientists." It applies to all computing professionals.

Research Ethics:



Research Ethics & Integrity:

- (1) Note, Research Integrity includes but is broader than research ethics considered here
 1. E.g. not falsifying or misrepresenting results, recognizing individual's contributions to research, responsible use of public funds, reported problems, ...
- (2) All UK research funders require consideration of Research Ethics.
- (3) Chinese legislation requires researchers in China to take account of Research Ethics

考点是学生是否意识到:

Research must comply with legal and ethical standards simultaneously.

Code of Research Conduct and Research Ethics – summary

1. It covers a lot of ground, including:
 - (1) Data, data management, especially personal data
 1. Including compliance with GDPR/Data

- 2. Protection Act
- (2) Insurance
- (3) International research
- (4) Research involving human participants
 - 1. Including but not only medical research
- (5) Research involving animals
- (6) Health and safety

Considerations for Research Participant Information

- 1. The research
 - (1) Aims and objectives of the research
 - (2) Funder
 - (3) Governance
- 2. Taking part (explain)
- 3. Risks and mitigation
- 4. Data processing
- 5. Storage and retention
- 6. Third party recipients, services, transfers
- 7. Legal rights and authorship
- 8. Ethical rights
- 9. Consent to participate

考试常用句:

Informed consent requires that participants understand the purpose, risks, and data usage of the research.

Principles of **research integrity**:

- 1. Research integrity means conducting research in a way which allows others to have trust and confidence in the methods used and the findings that result from this.
- 2. Within the University, conducting research with integrity also means meeting the professional standards expected of our researchers.
- 3. The scientific enterprise is built on a foundation of trust. Society trusts that scientific research results are an honest and accurate reflection of a researcher's work. Researchers equally trust that their colleagues have gathered data carefully, have used appropriate analytic and statistical techniques, have reported their results accurately, and have treated the work of other researchers with respect.
- 4. Research integrity may be defined as**
 - (1) Active adherence to the ethical principles
 - (2) And professional standards
 - (3) Essential for the responsible practice of research.

In the definition:

- 1. By **active adherence** we mean adoption of the principles and practices as a **personal credo**, not simply accepting them as impositions by rulemakers.

2. By **ethical principles** we mean **honesty, the golden rule, trustworthiness, and high regard for the scientific record.**

金句总结:

Research integrity is about trust in methods and results.

Issues and principles of research integrity(核心原则):

- Honesty and Truthfulness
- Objectivity
- Data Transparency
- Informed Consent
- Confidentiality
- Plagiarism Avoidance
- Authorship and Contributorship
- Conflicts of Interest
- Research Design and Methodology
- Reproducibility and Replicability
- Peer Review

考试高频:

falsification / misrepresentation = serious ethical breach

Responsible Research and Innovation(RRI)

The AREA Framework:

1. Anticipate – describing and analysing the impacts that might arise.
2. Reflect – reflecting on the purposes of, motivations for and potential implications of the research.
3. Engage – opening up such visions, impacts and questioning to broader deliberation, dialogue, engagement.
4. Act – using these processes to influence the direction and trajectory of the research and innovation process itself.

考试金句:

Responsible innovation requires ongoing reflection and adaptation, not just compliance at the start.

RRI – The 4Ps:

1. Process: covers all activities in preparing research, undertaking data collection and analysis, storage and presentation of data and interaction with respondents.
2. Product: can refer to products or services. It includes the consequences of use as well as misuse of research products and the impact that research has on the natural and social environment.
3. Purpose: covers the question why research is undertaken at all.

4. People: are at the heart of RRI and need to be explicitly considered.

加分点:

People are at the heart of responsible research and innovation.

Limits of RRI:

1. RRI cannot:
 - (1) Predict the future
 - (2) Avoid all problems arising from research, innovation and technology development
 - (3) Eliminate value conflicts
 - (4) Relieve researchers / funders / industry from responsibility
 - (5) Be a panacea
2. RRI can:
 - (1) Stimulate an intelligent conversation about R&I(Research & Innovation)
 - (2) Facilitate second order reflexivity

考试万能句;

RRI does not eliminate ethical dilemmas but provides a structured way to address them.

万能考试句:

1. **Research ethics concerns whether and how research should be conducted.**
2. **Research integrity is essential for trust in scientific findings.**
3. **Informed consent is a fundamental requirement in research involving humans.**
4. **Responsible research and innovation requires anticipating and reflecting on societal impacts.**
5. **Ethical responsibility extends beyond legal compliance.**

Lec 6

Equity – Diversity – Inclusion

1. Equity:

Give everyone an even opportunity to participate and achieve.

Equity ≠ Equality

Equity 关注 outcomes (结果), 而不仅是 inputs (规则)

Equity: different support based on needs → fair outcomes, not identical treatment

Equality: everyone treated the same → same rules, same resources

考试标准结论:

Treating everyone the same does not necessarily result in fairness.

2. Diversity:

Encourage and celebrate differences between people.(**Diversity ≠ tokenism**)

Encouraging and celebrating participation by people from wide range of backgrounds and with great range of life experiences.

3. Inclusion:

Everyone is included and belongs.

Inclusion means that everyone is made welcome and feel they belong and can participate.

Everyone is treated fairly and with respect, regardless of their differences.

重点: 能不能真正参与、是否被尊重

Protected Groups:

There are some groups that require special consideration for EDI and may experience discrimination and hence require protection:

(1) **Age**

(2) **Disability (including physical, mental health, autism...)**

(3) **Sex: Male / Female**

(4) **Race**

(5) **Religion and belief**

(6) **Sexual orientation**

Intersectionality:

It is important to consider protected groups; **But** sometimes discrimination happens at the **intersection between groups. (For example, black women in engineering.)**

考试加分句:

Ethical analysis should consider intersectionality, as disadvantages often compound across multiple protected characteristics.

Action on EDI

1. Awareness / vigilance
2. Remedial action (e.g. improving conditions for disabled)
3. Positive action (e.g. encouraging diversity)
4. Self-reflection on unconscious bias
5. UNNC strategy and policy on EDI

EDI and Engineering:

- **Equity, Diversity and Inclusion are becoming increasingly important in Engineering.**

- **EDI in the workplace:**

- (1) Diverse workplaces increase the skill set of the team.
- (2) "The most diverse companies are now more likely than ever to outperform less diverse peers on profitability."
- (3) Clear and fair treatment of all employees makes the workplace a more attractive and positive place to work.
- (4) Diversity of employees leads to better decisions.

- **EDI in engineering, design, development and innovation:**

- (1) Embracing EDI into our engineering solutions, so that everyone can enjoy and participate in our products and technologies.
- (2) When designing, think about groups who may have difficulty using it. ... Awareness
- (3) Create focus groups and talk with people who may need special arrangements to use your product or design.

高频考句:

Inclusive design leads to better and more ethical technological solutions.

Accessibility in Software:

1. Provide options for colors, font, and font size.
2. Narration tools.
3. Magnifier.
4. Subtitles.

考试高频结论:

Failure to consider accessibility can result in indirect discrimination.

在 PEC 中:

- **Fairness $\neq$ Equality**

- Fairness 更接近:
 - Equity
 - outcomes-based fairness
 - attention to vulnerable groups

考试标准回答:

The scenario adopts a particular definition of fairness. Rather than simple equality, fairness here should be understood in terms of equity, which focuses on ensuring fair outcomes by recognising different needs and vulnerabilities.

万能考试句:

1. Fairness is not the same as treating everyone equally.
2. Equity focuses on fairness of outcomes rather than identical treatment.
3. Protected groups may require special consideration to avoid discrimination.
4. Inclusion concerns whether people are able to meaningfully participate.
5. Accessibility is a core ethical requirement in engineering and computing design.

Lec 7

Learning Outcomes:

Perspectives on Privacy:

- Dimensions of privacy
- Threats and principles
- Trade-offs and rights

Theories of Privacy:

- Definitions and views
- Solove's Taxonomy

Different Opinions on Privacy

Perspectives on Privacy:

必考：怎么“看”隐私：

1. Expectations of Privacy
“public system” $\neq$ “public content”

考试金句：

The ease of accessing information does not eliminate reasonable expectations of privacy.

2. Consent

考试成熟表达：

Consent obtained under conditions of power imbalance may not be ethically sufficient.

A list of privacy risks & threats(高频):

- A. Intentional use or release
- B. Unauthorised use or release
- C. Inadvertent leakage or careless loss
- D. Search query data
- E. Re-identification of individuals
- F. Smartphone data
- G. Massive data storage (i.e., the cloud)
- H. Complexity of software
- I. Surveillance and recognition
- J. Unintended use
- K. Difficult to control or protect (e.g. autonomous systems such as AI).

必考结论：

The existence of data itself presents a risk.

可直接写：

Even when no misuse occurs, the mere existence of large data sets creates privacy risks.

Theories of Privacy:

Definitions:

(1) Three dictionary definitions of privacy:

1. **Seclusion (being set apart, or out of view)**
2. **Secrecy or concealment**
3. **Freedom from intrusion**

Warren & Brandeis's argument:

1. Opponents state that "privacy" derives from other rights(反驳常见的说法):

(1) **Implicit contract**(隐私=合同问题):

(2) **Intellectual property**(隐私=知识产权问题):

隐私不是这些权利的附属物

2. Warren & Brandeis debunked this, i.e., "Privacy" is not protected by contracts and intellectual property:

考试可用句：

Warren and Brandeis argue that privacy is an independent right, not merely derived from contract or property rights.

Solove's Taxonomy of Privacy

Four categories of privacy violation:

1. Information collection

(1) **Surveillance:**

Monitoring continuously, usually via audio, visual, or computer technology.

(2) **Interrogation:**

"Pressuring... individuals to divulge information"

2. Information processing

(1) **Aggregation:**

Collecting many small pieces of information about a person and linking them together, to create new information.

(2) **Identification:**

"Connecting information to individuals".

(3) **Insecurity:**

Inadequately safeguarding collections of personal data against theft.

(4) **Secondary Use:**

Using data that people willingly gave for one purpose for some other purpose they did not approve.

(5) **Exclusion:**

Failing to notify individuals that their data is being collected, or failing to provide a way for individuals to view or correct such data.

3. Information dissemination

(1) Breach of Confidentiality:

Breaking a contractual ... duty to keep someone else's information private.

(2) Disclosure:

Publishing private, but true, information in a way that damages the reputation of the subject.

(3) Exposure:

Publicly displaying certain physical or emotional attributes of another that are normally considered private, especially if such as display is humiliating or embarrassing.

(4) Increased Accessibility:

Making records that are technically available to the public easier to access.

(5) Blackmail:

Exerting power over another by threatening to reveal damaging information about that person.

(6) Appropriation:

Using someone else's identity for one's own ends.

(7) Distortion:

"Manipulat[ing] ... the way a person is perceived or judged by others".

4. Invasion

(1) Intrusion:

Communicating with people in a way that disturbs their peace or makes them feel uncomfortable.

(2) Decisional Interference:

Intruding into an individual's decision making regarding their private affairs.

考试万能句:

This scenario constitutes a privacy violation under Solove's taxonomy, particularly in the category of information processing through secondary use.

Different Opinions on Privacy

1. The "Nothing to Hide" argument:

Richard Posner believes **privacy as a social good is greatly overrated.**

考试反驳句 (非常好用):

The "nothing to hide" argument fails to recognise the cumulative and systemic harms of data aggregation and misuse.

2. Juan Enriquez on "Your Online Life, Permanent as a Tattoo"

3. Alessandro Acquisti on "What Will a Future Without Secrets Look Like?"

4. Glenn Greenwald on "Why Privacy Matters?"

考试万能句:

1. Privacy is not eliminated simply because information is accessible.

2. The existence of personal data itself creates ethical risks.
3. Privacy is a distinct right, not merely derived from contracts or property.
4. Solove's taxonomy provides a practical framework for analysing privacy violations.

5. The "nothing to hide" argument overlooks systemic and cumulative harms.

Lec 8

Intellectual & Intangible Property

1. **The products of creation and invention** (e.g., a book, a musical composition, computer software...) **are typically called "intellectual" property** to distinguish them from **tangible property** (e.g., a house, a car, a pair of shoes...)
2. Today, we look at:
 - (1) **Types of intellectual property** that are most affected by **computing and the Internet**
 - (2) **New types of intangible property** that arise due to computing, but **do not seem creative** in nature

关键理解:

IP 保护的不是“想法本身”，而是 **想法的表达或实现**

直接连到 Jefferson 的名言 (page 7):

Ideas are naturally non-rivalrous once disclosed.

The basic idea / argument:

1. The importance of creativity and invention:
Protecting innovators and their creation
2. Ultimately to encourage innovation
But does the opposite happen?
3. Defining intellectual and intangible property
Intellectual property often shortened to IP

考试常见反问:

Does IP law sometimes hinder innovation instead of encouraging it?

正确态度:

这是一个持续争论的问题 (experts disagree)

This is an issue that is still under debate (experts disagree).

Intellectual Property (IP) Law:

1. To ensure that **people(creators)** with good ideas can **profit from them:**
even once the idea is known to everyone
2. To **encourage** people to get involved in **creative arts and invention:**
for the benefit of everyone
3. IP Law is very complicated
4. We focus on **three key aspects** that directly affect computing:
 - (1) **Copyright**
 - (2) **Patent**

(3) Trademark

IP Protection: 5 Key Questions (遇到 IP 情景题, 用这 5 问直接拆题)

- For each type of IP protection, we ask:
 1. What type of thing can be protected?
 2. What rights are reserved for the creator of the work?
 3. What rights are reserved for the public?
 4. How does one obtain protection for a work?
 5. How long does the protection last?

Copyright:

1. Main mechanism for **protecting creative works**, e.g., art, music, & literature
2. US Law **DOES PROTECT**:
 - “**Original works of authorship fixed in any tangible medium of expression**” (Title 17, US Code section 102) in the areas of literature, music, drama, pantomime, graphic art, sculpture, motion pictures, sound recordings, and architecture
3. US Law **DOES NOT PROTECT**:
 - Ideas, facts, or common knowledge
 - Creative works that are **not** in a “tangible” fixed form
 - E.g., dance choreography: to protect work, choreographers often make video recordings of dance during rehearsal
4. **No “process” to obtain copyright**
 - works created are **automatically protected once they appear in a fixed medium**
5. Duration – (Currently) **Lifetime of the author + 70 years**
 - When Old, Out-of-Copyright Art Becomes **Free Art**.

Four main rights for the author:

1. The right to **reproduce** the copyrighted work
2. The right to **prepare derivative works** based on the copyrighted work
3. The right to **distribute copies of the copyrighted work** to the public by **sale or other transfer of ownership**, or by rental, lease, or lending
4. The right to **perform or display** the copyrighted work publicly

Rights for the public:

1. Most activities not covered by four main rights of the author belong to the public.
 - E.g., if you **own a DVD**, you have **right to display** privately in own home
2. There are **two main rights reserved for public** that deserve special attention:

(1) Fair Use

1. We can, in certain situations, use parts of copyrighted work without express permission
2. Following are **not an infringement of copyright**:
 - Criticism
 - Comment
 - News Reporting

- Teaching (inc. multiple copies for classroom)
 - Research
3. Factors of Fair Use:
- Law does not give clear rules
 - Four factors considered by **court /arbitrator for each case:**
 1. **Purpose and character of use:**
 - e.g., commercial vs. non-profit educational
 - **noncommercial and educational more likely to be fair**
 2. **Nature of the copyrighted work**
 - e.g., fiction vs. nonfiction, published vs. non-published
 - **use of nonfiction work & published work more likely to be fair**
 3. **Amount and substantiality of used material**
 - relative to the copyrighted work as a whole
 - **short extracts more likely to be fair**, but not if substantial
 - **少 ≠ 一定 fair (关键部分就不行)**
 4. **Effect of use on potential market** for, or value of, the copyrighted work
 - use more likely fair if copyright holder's ability to **make money is not damaged**

Not fair use:

- (1) When using the work, it is a vital or recognizable part, and therefore substantial, it will cause unfair use

常见误区: 这部分需要具体情况具体分析, 没有一定就是对的, 合理的可能

✗ "educational = fair use"

✗ "Only a small part = fair use"

考试金句:

Fair use is context-dependent and must be evaluated case by case.

(2) First Sale

Doctrine of first sale: **No second royalty**

1. **Copyright holders are not entitled to a second royalty from resale**
(消费者买了实体的副本可以转卖但不可复制)

Digital Rights Management (DRM)

1. Technologies that work to ensure copyrighted content can **only be viewed by the purchaser**
2. When you purchase content **it is locked to your device**
3. ebook purchased for your Amazon Kindle cannot be read on your friend's Kindle

What happens to Fair Use and Fair Sale rights? (e-book with DRM):

1. You may not be able to **cut and paste a quote** from a book [can't exercise right to fair use]

2. You cannot give copy to friend and then delete your copy [can't exercise right to resell a book – i.e., **doesn't meet first sale doctrine**]

DRM 导致: Fair use 和 first sale 被技术手段侵蚀

Plagiarism vs. Copyright Violation

1. Posner's definition of plagiarism
 - **copying that the copier claims is original with him** and the claim causes the copier's audience to behave otherwise that it would if it knew the truth
(抄袭: 声称作品是自己的)

- **Plagiarism ≠ Copyright infringement**
- **Copyright infringement ≠ Plagiarism**

考试高频例子:

有授权但冒充原创 → plagiarism

给出处但无授权 → copyright violation

Sharing vs. Selling

Patent Law

Definitions:

1. Intellectual works of invention, **value is on its usefulness**
 - Covers processes, machines, manufacture, compositions of matter, ...
 - Capable of being made or used
2. **Need to apply for it**
 - Can be **expensive process** (more than \$10,000)
3. Criteria, invention must be:
 - **Novel** (not been invented before)
 - **Non-obvious** (not obvious to another specialist in appropriate area)
 - Valid for some fixed period of time: (usually **20 years** from application date)
4. **Good investment!**
 - No-one can make, use, offer to sell, or sell the invention without the patent holder's permission!

Trademark

Definitions:

1. Legal registration of word, phrase, symbol, or item that **identifies** product, service or corporation
2. Usually used to **protect word or logo related to a product**, so that consumers can be **sure products are authentic**
3. Can register trademark for **identifying characters of brand**
 - **Colors:**
 - T-Mobile holds trademark for colour magenta

- **Sounds:**
Motorola holds trademark on the “Chirp” sound phones make
 - **Shapes:**
Coca-cola holds trademark on “bottles, jars, or flasks with bulging, protruding or rounded sides”
4. **Trademark vs. Brand:**
 - (1) **Brand** refers to the product's identity **as a whole**
 - (2) **Trademark** refers to **specific element of identity elements**
 5. **In order to receive TM**, you must **already** be using the mark to represent product or service
 6. Can register for **fee of few hundred dollars**
 7. TM lasts **10 years** but **can be renewed indefinitely**
 8. As long as **you own trademark**, **no one else can** use it in a way that is likely to **cause confusion among customers**
 - no other mobile phone company can adopt magenta as company colour, but a soft-drink can could use magenta
 9. **Trade dress—look and feel** of a product and its packaging
 - Several companies have attempted to use trade dress protection to prevent others from copying look and feel of their Web sites

考试易错点：

1. trademark ≠ brand
2. trademark 保护的是 标识元素

Trade Secrets

1. With some products **it is possible to keep the intellectual “idea” secret**
 - an Internet company's search algorithm
 - Coca Cola's secret recipe
2. Such secrets are called “trade secrets”
 - They are secrets that are fundamental to business
3. IP laws are not needed to protect trade secrets
 - If somebody breaks into company and **steals the secret**, they can be **prosecuted for trespassing**
 - Employees can be prevented from **leaking secrets** through **use of non disclosure agreements** in their employment contracts
4. **IP laws needed only for inventions/creations that are no longer secret**

IP Protection for Software

1. **Copyright, Patents, and Trademark can be used to protect parts of a computer software product**
 - source code logic: Source code is a **creative work** of authorship and is **copyrighted**
 - look and feel: **Copyright/Trademark**
 - features: **Patent**
 - algorithms: **Patent**

2. It is not a simple question of how to protect software

Notes of Software Copyright:

1. Work can be licensed to others
2. Copyright can be sold or assigned to others
3. Claims for infringement must be proven in court
4. Copyright in work done as part of a job belongs to the employer
5. All digital media are likely to be subject to copyright
6. Creative commons licenses are like open source licenses for digital media rather than software

Notes on Patents for Software

- Complex, contentious area
- Legal protection from others exploiting the invention
- Use of invention can be licensed to others
- Typically protected for up to 20 years
- The essence of the invention is disclosed and published
- Claims for infringement must be proven in court
- But there is no agreement

Summary

- Qu: Which type of intellectual property protection should apply to software?
 - This is a **trap!**
 - There is **no clear or obvious answer**
 - The **experts disagree**
- Possible solution – a new legal framework
 - **Difficult to map existing law**, designed for tangible goods onto intangible ones

IP 保护类型	版权 Copyright ©	专利 Patent	商标 Trademark ™
可以保护什么?	创意作品 (艺术、音乐、戏剧、文学等), 必须以固定且可感知的形式存在	发明 (流程、机器、制造方法等), 必须能够被制造或使用	识别产品、服务或公司的文字、短语、符号、标志、颜色、声音、形状等
创作者拥有的权利?	1. 复制权 (reproduce) 2. 衍生作品权 (改编) 3. 分发复制品 4. 公开表演或展示	独占权: 制造、使用、销售发明 (或授权他人)	独占使用商标的权利 (防止他人使用可能导致混淆的标志)
公众的权利?	除作者权利以外, 还包括: 1. 合理使用 (Fair Use) 2. 首次销售原则 (First Sale: 可转卖作品)	未经许可不得制造、使用、提供销售或销售相关发明	不得以可能造成消费者混淆的方式使用商标
如何获得保护?	自动获得, 一旦作品以固定形式出现 (写下、录下、保存)。费用: 免费	必须向专利局申请并满足条件: • 新颖性 (novel) • 非显而易见性 (non-obvious) 费用: 通常 1 万美元以上	必须注册商标, 并且在注册前必须已经用于商品或服务。费用: 几百美元
保护期限?	作者终身 + 70 年 (当前法规)	通常为自申请日起 20 年	每 10 年可续期一次, 可无限期续期 (必须持续使用)

考试万能句:

1. Intellectual property law aims to balance innovation and public interest.
2. Copyright protects expressions, not ideas.
3. Fair use must be evaluated case by case using four factors.

4. Plagiarism and copyright infringement are distinct concepts.
5. There is no single correct way to protect software using existing IP law.

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Trust

In some cases, we are right to trust computers, in other cases we are wrong

考试万能句:

Trust in computing systems is context-dependent and can be misplaced when users overestimate system reliability.

Causes of Computer Failure(具体说出哪一类别)

1. Hardware errors

- Hardware malfunctions causing erroneous results

(Example: Intel Pentium FDIV Bug → Even rare errors can be ethically significant when systems are widely deployed.)

2. Software errors

- Software “bugs” in code or software design

(Example: Ariane 5 → This was a preventable software error exacerbated by inadequate error handling and risk assessment.)

3. Computer solves the wrong problem

- Programmers incorrectly understand requirements of clients. Program does as intended (no bugs) but does not do what client wants and expects

4. Misuse

- Using system for the wrong purpose, or using system incorrectly for the right purpose. E.g., “Garbage in, garbage out”

5. Communication failure (in human interaction)

- User prompted for action misunderstands (e.g., airplane refuel, user assumes amount in gallons rather than litres – plane runs out of fuel!)

6. Malice

- Malicious software (virus, trojan, hacking ...) for profit, terrorism, warfare, “fun”

Safety-Critical Software (SCS)

1. Software that **affect somebody’s safety** if fails to work properly

- Air traffic control, nuclear safety system, medical, etc.

2. Standards of creating, testing, and maintaining SCS are much **stricter** than other software

3. **Exhaustive testing is the obvious answer**, but **large number of decision-points** in software **makes this impossible**

4. Software also **control machinery and equipment**, therefore

- Real-time software systems (e.g., avoid object before crash)
- Software executes in multi-process environments (e.g., accelerate, brake, avoid)
- These greatly complicate testing

考试结论：

Safety-critical systems require much stricter standards of design, testing, and maintenance.

Towards Better Safety-critical Software

1. **Better education and training** of engineers, programmers, analysts
2. **More sophisticated software testing procedures**
 - May include formal verification (like a mathematical proof)
3. **Better maintenance**
4. **Refocus on prevention** rather than detection/correction
5. **Simulation** can be beneficial (e.g., air-disaster)
6. **Emphasis on ethics and professional education**

考试可用句：

Preventative approaches and professional training are essential to reducing the likelihood of catastrophic failure.

Therac-25:

- 医疗安全关键系统
- 过度依赖软件替代硬件保护
- 缺乏独立代码审查、风险建模
- 错误信息设计导致操作员误用
- 结果：多起严重伤害与死亡

关键总结：

This was not a single bug, but a system engineering failure.

Accountability & Moral Responsibility

1. The following people bear moral responsibility
 - (1) The programmer who designed software
 - (2) The company's quality assurance officer
2. Not legal responsibility
 - That is a matter for the courts to decide

Birsch's Moral responsibility

3 criteria must be met for person to bear moral responsibility:

- (1) "The person's actions must have **caused the harm**, or been **a significant causal factor**
- (2) The person must have **intended or willed the harm**, or it must be a result of his/her **negligence, carelessness, or recklessness**
- (3) The person must **be able to have known**, or **must know the consequences of the action**, or must have **deliberately remained ignorant of them**"

考试标准结论句：

According to Birsch's criteria, the programmer bears moral responsibility due to foreseeable risks and professional negligence.

Malware

1. Worm

- makes copies of itself and propagates through network to infect other computers

2. Virus

- similar to worm but resides in another program. Program must execute for virus to propagate

3. Spyware

- program secretly installed for purpose of collecting information about computer's user(s)

4. Trojan Horse

- software masquerades as innocent and perhaps useful, but actual malicious

5. Rootkit

- program that embeds into OS and acquires special privileges. Potential to override authority of OS, gaining access to passwords and all files

考试陷阱:

Rootkit ≠ virus (权限层级不同)

“Ethical” Hacking

1. “Hackers”: people who write and deploy malware

2. “Malware” indicates malice, so directly immoral

- To vandalize other's property, steal from their credit card, or spy on a person, etc. is clearly immoral

3. Some “white hat” hackers have legitimate reasons

- To seek vulnerabilities in system before malicious attacker finds them
- Sometimes ethical hacking violates the law: you need legal authority
- Education & training of ethical hackers is controversial

4. Zero-day attack

- Attack that exploits previously unknown vulnerability

5. Humans often a weak link in system security

- Phishing, tricking users, listening devices & cameras

成熟表述:

Ethical hacking may be morally justified in some contexts but still requires legal authorisation and strict safeguards.

考试万能句:

- 1. Trust in computing systems is often justified but can be misplaced.**
- 2. Even rare system failures can have serious ethical consequences.**
- 3. Safety-critical software requires higher professional standards.**

4. **Moral responsibility depends on causation, negligence, and foreseeability.**
5. **System failures are often the result of organisational and engineering failures rather than a single bug.**

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AI Risks

We can divide AI risks into **5 broad categories**:

1. **Data related:**

Data is an important resource for AI; poor quality or misuse can create problems.

(1) **Poor Data Quality**

Data source may be unknown or unreliable. Measured data may not be accurate or wrongly entered.

(2) **Misuse of Data**

Even if data is reliable, the data may be used without proper permission or in an inappropriate way, that differs from the intended reason for its collection.

(3) **Data Bias**

Collected data may present a bias related to how it is collected, and/or for protected classes.

(4) **Poor Security**

Data may be used correctly but may not be stored, anonymized or communicated securely within the AI system.

2. **Performance:**

Whether AI performs to benefit all its stakeholders, or not.

(1) **Poor Aggregate Performance**

The AI may not perform well at the assigned task, in general (poor model fit). It may generally give biased outcomes (poorly calibrated). It may be evident that more training data is required to improve performance.

(2) **Overfitting & Brittleness**

Machine learning-based AI uses training data to learn. Often it may perform extremely well on training data, but poorly on new data. This is called overfitting. Similarly, the AI may adapt poorly to changes in data (brittleness), e.g. over time or place.

(3) **Sub-population Performance Bias**

Even if aggregate performance is good, the AI may perform badly for a specific sub-population. This could be a protected class or a combination (intersectionalism).

(4) **Unfairness**

The AI may make decisions that are unfair to a particular group, in some sense. It turns out there are several different definitions of fairness, so the AI stakeholders and developers need to determine which applies.

关键句:

Good average performance does not guarantee fairness for all groups.

3. Human/AI interaction:

How humans interact with the AI presents a risk.

(1) Human-Computer Interface Design

Poor interface design may lead to improper use of the AI, or misunderstanding of its decision, or miscommunication of objectives.

(2) Opaque System

If the AI cannot explain its decisions, it means we may not be able to understand how it works, and may mean the decision or the reason for the decision is wrong and we would never know. We may also need to explain decisions to customers or other stakeholders.

(3) Over- or Under-reliance

End-users may over-rely on AI system, or have irrational mistrust of AI.

(4) Misuse of AI

The AI may be designed in a way that makes it available for misuse, either by the organization that develops it, or by external parties that make use of the AI.

4. Objective:

How we code up the objective for the AI can be problematic, with unintended consequences.

(1) Ethical misalignment

The AI may not correctly embody ethical principles and hence may behave in an unethical way.

(2) Misaligned objective

Converting a complex requirement to formulae may lead to errors, or unintended consequences

(3) Gaming the objectives

The AI may find a way to meet an objective, but in a way that is inappropriate. (e.g. easiest way to prevent a car crash is to ensure the car does not move!).

考试句:

Encoding complex ethical goals into objective functions can lead to unintended consequences.

5. Social risks:

Broad social impact of AI, in general.

(1) Lack of expertise in AI.

(2) Singularity Risk.

(3) Human behaviour modification (e.g. loss of autonomy; cyborgs).

(4) Reflexivity (AI's influence on itself).

(5) Risk to the economy and jobs.

(6) Exploitation of human workers in AI development.

(7) Climate Risk.

AI Assurance

Broad principles to mitigate AI Risk:

1. Society will benefit from regulation and governance of AI.
2. Training and professional accreditation of AI professionals.
3. Well-established methods for AI development, validation and deployment.
4. Validation and monitoring of AI systems.
5. Understanding the conditions when we should not use AI.
6. Clear responsibility, accountability and governance of AI.

考试万能总结句:

Trustworthy AI requires governance, technical robustness, and human oversight.

Autonomous Agents

1. AI is deployed as decision support or as autonomous agent:
2. Just like human decision making, AI decision making can have ethical consequences.
3. There is human oversight over decision support AI, so ethics can be monitored and decisions over-ridden (at least, in principle).
4. However, **autonomous agents must make their own ethical decisions.**

Prof. Stuart Russell's general ethical principles (2017):

1. **Human-focused.** The AI's only objective is to maximize the realization of human values.
2. **Humility.** The AI is initially uncertain about what those values are.
3. **Learning.** Human behavior provides information about human values.

Fairness: Protected groups

1. In society, unfairness is linked to discrimination.
2. When thinking about fairness and discrimination, there are **special categories** that, in particular, we look to protect
 - (1) Gender
 - (2) Ethnicity
 - (3) Religion
 - (4) Age
 - (5) Marital Status
 - (6) Citizenship

Decision Support Systems:

- A = the sensitive attribute we wish to check for fairness [(如 gender)]
- R = a score or decision from the AI about an individual. [AI 决策 / 评分]
- Y = the outcome the AI is modelling [真实结果]

Fairness and AI:

1. Independence

- The AI decision should be **independent** of any individual characteristic.
- $R \perp A$ means R (score/decision) is independent of A .

In terms of probabilities, for any two values a, b of A :

$$P(R = r | A = a) = P(R = r | A = b) \rightarrow \text{conditional selection rate: 汽车保险}$$

2. Separation

- Same as Independence but the independence is within each outcome type being modelled.
- $R \perp A | Y$

In terms of probabilities, for any two values a, b of A :

conditional false positive rate (FPR)

$$P(R = r | A = a, Y = 0) = P(R = r | A = b, Y = 0)$$

conditional true positive rate (TPR)

$$P(R = r | A = a, Y = 1) = P(R = r | A = b, Y = 1)$$

This is a relaxation of the definition and allows some difference between groups between different outcomes

3. Sufficiency

- The AI decision should be sufficient to **account for all features and characteristics**
- This definition states that **the AI decision (R) should be sufficient to account for all variables (A), including the sensitive attribute.**
- For any r, a :

$$P(Y = 1 | R = r) = P(Y = 1 | R = r, A = a)$$

(conditional) positive prediction value (PPV): $R=1$

(conditional) false omission rate (FOR): $R=0$

4. Approximations

- In practice, there will not be an exact match between measured probabilities on data.
- Hence, **approximations** may be used.
- For example, for Sufficiency,

$$P(R = r | A = a) \approx P(R = r | A = b)$$

- 5. However, each of these definitions are different and generally to achieve one means discarding the other two.

必考结论句 (请背):

These fairness definitions are mutually incompatible; achieving one often requires sacrificing others.

考试标准回答方式: → 回答 what definitions of fairness → 代表一种书序约束

The scenario implicitly adopts a particular definition of fairness. In social terms, fairness aligns with equity, whereas in AI systems it is often formalized through statistical definitions such as independence, separation, or sufficiency.

Brittleness & Artificial Stupidity

考试句:

High performance on benchmark data does not guarantee robust real-world behavior.

Explainable AI:

1. Black Box

- (1) Often AI systems are too complex for us to understand how they are working and making decisions.

- (2) Hence, we call them Black Boxes.
- (3) Examples include Deep Neural Networks, Random Forests, Gradient Boosted Trees, and Support Vector Machines.

2. Explainable AI Metrics;

(1) LIME.

- **Local Interpretable** Model-agnostic Explanations
- For a given input x and AI model f that outputs a decision $f(x) = \hat{y}$, build an approximate local linear model around $f(x)$.
- Report the coefficients of this local model.
Positive coefficients mean positive contribution to $\hat{y}$
Negative coefficients mean negative contribution to $\hat{y}$
- Example: Bike rental dataset

Warmer temperature and good weather situation have a **positive effect** on the prediction.

The x-axis shows the feature effect: The weight times the actual feature value.

(2) SHAP.

- SHapley Additive explanation
- **Feature contributions**

(3) PDP.

- Partial Dependency Plots
- Computes marginal effect of a feature across a range of values:
 $PDP(x) = \mathbb{E}_w(m(x, w))$, where m is the model, x is value of feature of interest, w are all other features used in model m .
- Also, can be 2-way to show interaction between two features:
 $PDP(x_1, x_2) = \mathbb{E}_w(m(x_1, x_2, w))$

方法	类型	能解释什么
LIME	Local	单个决策“为什么”
SHAP	Local + Global	特征贡献
PDP	Global	特征整体趋势

考试句:

Explainable AI techniques help mitigate opacity and support accountability, but do not eliminate ethical risks.

Trusting AI:

1. Society will benefit from regulation and governance of AI.

2. Training and professional accreditation of AI professionals.
3. Well-established methods for AI development, validation and deployment.
4. Validation and AI monitoring.
5. Knowledge of conditions when we should not use AI.
6. We need to integrate into the AI development pipeline
 - (1) AI Risk Measurement,
 - (2) AI Risk Mitigation
 - (3) AI Risk Monitoring

考试万能句：

1. Trustworthy AI requires addressing technical, social, and ethical risks.
2. AI fairness has multiple formal definitions that are often incompatible.
3. Good average performance does not ensure fairness for all sub-populations.
4. Autonomous systems raise complex questions of moral responsibility.
5. Explainable AI supports trust and accountability but cannot resolve all ethical concerns.

高频问题：

Who should be held accountable?

Moral responsibility may be distributed across multiple actors, including system designers, developers, management, and organisations. While AI systems lack moral agency, human actors retain responsibility due to design choices, foreseeable risks, and professional obligations.

并且 自动联动：

- Birsch criteria
- Due diligence
- Organisational failure

Should we use AI at all in this scenario?

In high-risk contexts where transparency, accountability, or meaningful human oversight cannot be guaranteed, the ethical response may be to refrain from deploying AI altogether.

这句话 在 Lec9 + Lec10 综合题里非常加分。